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Point Mass Planning and NMPC Tracking for Time-Optimal Agile Quadrotor Gate Navigation

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ABSTRACT Time-optimal agile flight through a sequence of three-dimensional gates requires a consistent coupling between trajectory planning and tracking control. Point-mass planners offer fast numerical solutions for free-final-time optimisation, but the value of progressively enriching the reference fed to a nonlinear model predictive control (nmpc) tracker through a differential flatness bridge has not been clearly characterised on multi-gate circuits. This paper presents a modular three-stage pipeline: an offline time-optimal point-mass model (pmm) planner solved with CasADi and IPOPT; a differential flatness bridge that maps the flat outputs and their time derivatives to a full quadrotor reference comprising desired position, velocity, attitude quaternion, body angular rate, and collective thrust; and a body-rate nmpc tracker implemented on the acados SQP-RTI scheme with an SO(3) logarithmic-map external cost. Two reference configurations are compared in a software-in-the-loop setup that drives the MuJoCo physics engine through ROS 2 under a randomly seeded external-force disturbance. The baseline, nmpc-Att, uses only a yaw-only attitude reference derived from the velocity heading and carries no angular-rate or thrust feedforward. The proposed configuration, nmpc-Full, activates the complete flat reference, including the full attitude quaternion, body angular rate, and thrust feedforward. Results on a seven-gate figure-8 and an eight-gate vertical loop show that reference enrichment yields a categorical improvement in gate crossing and a systematic reduction of tracking error without increasing solver cost. The loop circuit further highlights a structural limitation of the common practice yaw-only reference: it cannot represent inverted attitudes, and only the flatness-based reference resolves this case.
GitHub: https://github.com/bryansgue/Time_optimal_planing-NMPC.

INDEX TERMS Agile flight, differential flatness, gate navigation, model predictive control, nonlinear control, point mass model, quadrotor UAV, time-optimal planning.

I. INTRODUCTION

AGILE quadrotor flight through a sequence of three-dimensional gates is a challenging benchmark in autonomous aerial robotics. In state-of-the-art racing systems, gate navigation drives the vehicle near its actuation limits, with reported peak accelerations in excess of $4g$ and peak speeds above 60 km/h while respecting geometric passage constraints [1]–[3]. Model-based approaches decompose this task into a planner and a tracker, coupled through an intermediate reference trajectory [4]. A central design question in this coupling is how much of the planner output should be communicated to the tracker: a position-only reference is simple to compute, while a full flatness-derived reference requires inverting the differential flatness map but carries attitude, angular rate, and thrust information that the tracker would otherwise have to reconstruct from error feedback alone. The quantitative effect of this choice on gate-crossing performance has received limited study in multi-gate scenarios.

A. RELATED WORK

Time-optimal planning for agile flight has converged on two families. Polynomial-flatness approaches following Mellinger and Kumar [5] cast trajectory generation as a quadratic program over piecewise segments and recover attitude and angular-rate references from flat outputs, but they are smooth rather than time-optimal and rely on separate time allocation [6], [7]. Point-mass formulations relax rotational dynamics to expose the translational time-optimality structure: Mueller *et al.* [8] derived closed-form pmm primitives between full kinematic states; Foehn *et al.* [1] extended this to multi-waypoint flight via Complementary Progress Constraints solved with IPOPT and matched expert pilot times; and the TOGT planner of Qin *et al.* [9] incorporated explicit gate geometry and single-rotor thrust bounds, solving nine-gate circuits in seconds. Zhou *et al.* [10] tackle the same time-optimal problem directly on the full 13-D quadrotor dynamics and close the loop with a time-adaptive MPC that adjusts the tracking horizon online, at the cost of coupling planning and control into a single monolithic formulation. Related minimum-time formulations extend to cluttered environments [11]. Across this family, trajectory quality is high but the tracking problem is left open.

Learning-based methods close the tracking loop by sidestepping the reference entirely. Loquercio *et al.* [12] trained a deep policy for 40 km/h flight in unstructured environments; the Swift system [3] reached world-champion-level racing using deep reinforcement learning; and Song *et al.* [4] showed that reinforcement learning can outperform optimal control by directly optimising the task-level objective, identifying the plan-then-track interface itself as a principal performance limit of model-based pipelines. The present work targets this interface from the opposite direction, enriching its content while preserving modularity.

Nonlinear model predictive control has become the dominant model-based tracker because the body-rate formulation [13] retains attitude-dependent coupling while delegating fast rate dynamics to an inner loop, and because real-time solvers such as acados with SQP-RTI [14], [15] deliver sub-millisecond QPs at 100 Hz. Robustness has been addressed by cascading nmpc with an L1 adaptive controller under unknown payload and wind [13], by augmenting with a Gaussian-process aerodynamic residual [16], and by incremental nonlinear dynamic inversion with flatness feedforward [17]; ROS-based MPC baselines for UAV trajectory tracking are widely available [18]. The benchmark comparison by Sun *et al.* [19] against differential-flatness-based control over 76 trajectories establishes nmpc superiority for dynamically infeasible references but fixes the reference structure and does not ablate its components.

Model Predictive Contouring Control fuses planning and tracking in a single online optimisation that maximises path progress while minimising contouring error [2], and recent variants add pmm-based online replanning around moving gates [20] and tunnel-based safety constraints that reach 100% crossing success above 80 km/h [21]; perception-latency bounds for high-speed flight are characterised by Falanga *et al.* [22]. MPCC performance is strong, but its planner and controller cannot be independently designed, validated, or replaced.

As a planning-control bridge, differential flatness was established for quadrotors in [5] and extended to rotor-drag models in [23], yielding algebraic maps from position and heading and their time derivatives to thrust, attitude, and body-rate references [24]. Greeff and Schoellig [25] integrated flatness feedforward into a linearised MPC, and Sun *et al.* [19] used flatness-derived references for both nmpc and DFBC without varying the reference components. A common practical compromise in nmpc design is to derive only a desired yaw angle from the trajectory planner ($\psi^* = \mathrm{atan2}(v_y, v_x)$) and convert it to a quaternion reference, while omitting the angular-rate and thrust feedforward that a full flatness inversion would provide. No prior work quantifies the performance gap between this common-practice incomplete reference and the complete flat output $(\boldsymbol{p}, \boldsymbol{v}, \boldsymbol{q}, \boldsymbol{\omega}, T)$ on a multi-gate 3D circuit.

Across these three families, three patterns recur: (i) PMM and TOGT planners deliver strong time-optimal references but do not specify how much of the flat output the tracker consumes; (ii) body-rate nmpc trackers have been benchmarked with fixed reference structures and without a gate-navigation objective; and (iii) MPCC fuses the two stages at the cost of modularity. No prior study decouples planner and tracker via a full flatness bridge and ablates the reference content on a multi-gate 3-D circuit under identical dynamics, solver, and cost; this is the gap addressed here.

B. IDENTIFIED GAP

Two related gaps motivate this study. *First*, a pipeline that combines an offline pmm planner, a differential flatness bridge, and a body-rate nmpc tracker receiving the full flat reference has not, to our knowledge, been reported as an integrated system in this specific form. Foehn *et al.* [1] combine pmm with MPCC tracking but without a flatness-derived full-state reference fed to an nmpc. TOGT [9] is a planner-only contribution. *Second*, existing studies of nmpc for agile flight typically fix the reference structure. The performance difference between the common-practice incomplete attitude reference (yaw derived from velocity direction) and the full flatness inversion has not been isolated on a multi-gate circuit under otherwise identical dynamics, solver, and cost structure. Identifying this difference precisely is necessary to justify the additional computation of the full flatness bridge.

C. CONTRIBUTIONS

The contributions of this paper are:

1) **Integrated pmm–flatness–nmpc pipeline.** A modular architecture coupling an offline pmm planner (CasADi + IPOPT), a differential flatness bridge that maps the flat outputs to a 17-component quadrotor reference, and a body-rate nmpc tracker (acados SQP-RTI, $N = 50$, 100 Hz) with an SO(3) logarithmic-map external cost.
2) **Flatness-inversion benefit on multi-gate circuits.** A MuJoCo software-in-the-loop evaluation on a figure-8 and a vertical loop with inverted apex shows that the angular-rate feedforward is the dominant factor for gate-crossing at aggressive rotational transitions, and that the tilt-corrected quaternion is necessary for the *feasibility*

of inverted segments that the yaw-only reference cannot represent by construction.

D. PAPER ORGANIZATION

Section II introduces the body-rate quadrotor model and the notation used throughout the paper. Section III formulates the time-optimal gate navigation problem and details the pmm planner. Section IV derives the differential flatness bridge. Section V formulates the nmpc tracker. Section VI defines the two reference configurations, describes the MuJoCo evaluation, and reports the results. Section VII concludes the paper.

II. QUADROTOR MODEL

The quadrotor is modelled at the body-rate level, delegating low-level attitude dynamics to an inner-loop rate controller, as is standard for agile nmpc [13]. The state $\boldsymbol{x} = [\boldsymbol{p}^\top, \boldsymbol{v}^\top, \boldsymbol{q}^\top, \boldsymbol{\omega}^\top]^\top \in \mathbb{R}^{13}$ collects the position $\boldsymbol{p}$ and linear velocity $\boldsymbol{v}$, expressed in the world frame $\mathcal{W}$ (with $\boldsymbol{e}_3 = [0, 0, 1]^\top$ aligned with gravity); the attitude quaternion $\boldsymbol{q} = [q_w, q_x, q_y, q_z]^\top \in \mathbb{S}^3$ in Hamilton convention, representing the rotation from the body frame $\mathcal{B}$ to $\mathcal{W}$ and inducing the rotation matrix $\mathbf{R}(\boldsymbol{q}) \in SO(3)$; and the body angular velocity $\boldsymbol{\omega}$ expressed in $\mathcal{B}$. The control input $\boldsymbol{u} = [T,\ \omega_x^c,\ \omega_y^c,\ \omega_z^c]^\top$ consists of the collective thrust $T \geq 0$ along $\boldsymbol{z}_\mathcal{B}$ and the commanded body angular rate $\boldsymbol{\omega}^c$. The continuous-time dynamics are

$$\dot{\boldsymbol{p}} = \boldsymbol{v}, \tag{1}$$

$$\dot{\boldsymbol{v}} = -g\,\boldsymbol{e}_3 + \frac{T}{m}\mathbf{R}(\boldsymbol{q})\,\boldsymbol{e}_3, \tag{2}$$

$$\dot{\boldsymbol{q}} = \tfrac{1}{2}\,\Xi(\boldsymbol{q})\,\boldsymbol{\omega}, \tag{3}$$

$$\dot{\boldsymbol{\omega}} = \frac{\boldsymbol{\omega}^c - \boldsymbol{\omega}}{\tau_{rc}}, \tag{4}$$

with vehicle mass m, gravitational acceleration g, and first-order rate-controller time constant τ_{rc}. The kinematic matrix $\Xi(\boldsymbol{q})$ in (3) is the standard Hamilton form [5]; the unit-norm constraint $\|\boldsymbol{q}\| = 1$ is preserved by renormalising $\boldsymbol{q}$ after each integration step. Model (1)–(4) serves as the prediction model of the nmpc (Section V) and as the substrate onto which the differential flatness map (Section IV) inverts the planner output.

III. POINT MASS TIME-OPTIMAL PLANNER

A. TIME-OPTIMAL GATE NAVIGATION PROBLEM

Let $\mathcal{G} = \{(\boldsymbol{g}_i, \hat{\boldsymbol{n}}_i, r_i)\}_{i=1}^{N_g}$ denote a sequence of N_g gates, each defined by centre position $\boldsymbol{g}_i \in \mathbb{R}^3$, unit normal $\hat{\boldsymbol{n}}_i$, and radius r_i. The time-optimal gate navigation problem seeks a trajectory that minimises the final time T_f subject to the quadrotor dynamics (1)–(4), actuator bounds on the collective thrust and body rate, and, for each gate $i = 1, \ldots, N_g$, a crossing instant $t_i \in [0, T_f]$ at which the vehicle pierces the gate plane within its radius:

$$\hat{\boldsymbol{n}}_i^\top(\boldsymbol{p}(t_i) - \boldsymbol{g}_i) = 0, \tag{5}$$

$$\|\boldsymbol{p}(t_i) - \boldsymbol{g}_i\|_\perp \leq r_i - \delta, \tag{6}$$

where $\|\cdot\|_\perp$ denotes the in-plane distance to the gate centre and $\delta > 0$ is a safety margin. Solving this problem directly on the full 13-dimensional model (1)–(4) with a free final time and N_g implicit crossing events is computationally challenging [1]. We therefore adopt a point-mass relaxation, which enables fast offline solution and still captures the translational structure that dominates gate-crossing timing; the flatness-based tracker of Section IV subsequently reintroduces the attitude and body-rate content that the relaxation discards.

B. POINT MASS MODEL

The pmm approximates the quadrotor as a point mass with state $\boldsymbol{s} = [\boldsymbol{p}^\top, \boldsymbol{v}^\top]^\top \in \mathbb{R}^6$ and control $\boldsymbol{a} \in \mathbb{R}^3$ (desired acceleration):

$$\dot{\boldsymbol{s}} = \begin{bmatrix} \boldsymbol{v} \\ \boldsymbol{a} \end{bmatrix}. \tag{7}$$

Attitude dynamics are abstracted away; this relaxation is valid when the rate-loop settling time τ_{rc} in (4) is small relative to the collocation step T_f/N used below.

C. NLP FORMULATION

The continuous problem is transcribed via direct multiple shooting with N_{PMM} collocation steps and free final time T_f. The variable step size $\Delta t = T_f/N_{\text{PMM}}$ couples the time horizon to T_f:

$$\min_{\boldsymbol{S},\boldsymbol{A},T_f}\ w_T\,T_f + w_a \sum_{k=0}^{N_{\text{PMM}}-1} \|\boldsymbol{a}_k\|^2\,\Delta t + w_j \sum_{k=1}^{N_{\text{PMM}}-1} \frac{\|\boldsymbol{a}_k - \boldsymbol{a}_{k-1}\|^2}{\Delta t} \tag{8a}$$

$$\text{s.t.}\quad \boldsymbol{s}_{k+1} = \text{RK4}(\boldsymbol{s}_k, \boldsymbol{a}_k, \Delta t), \tag{8b}$$

$$\boldsymbol{s}_0 = \boldsymbol{s}_{\text{init}}, \quad \boldsymbol{s}_{N_{\text{PMM}}} = \boldsymbol{s}_{\text{final}}, \tag{8c}$$

$$\hat{\boldsymbol{n}}_i^\top(\boldsymbol{p}_{k_i} - \boldsymbol{g}_i) = 0, \quad \|\boldsymbol{p}_{k_i} - \boldsymbol{g}_i\|_\perp \leq r_i - \delta, \tag{8d}$$

$$d_i^- < 0 < d_i^+, \quad i = 1, \ldots, N_g, \tag{8e}$$

$$\|\boldsymbol{v}_k\| \leq v_{\max}, \quad \|\boldsymbol{a}_k\| \leq a_{\max}, \quad T_f \geq T_{\min}, \tag{8f}$$

where $d_i^\pm = \hat{\boldsymbol{n}}_i^\top(\boldsymbol{p}_{k_i\pm1} - \boldsymbol{g}_i)$ enforce the correct approach direction through each gate (8e). The three terms in (8a) trade the final time T_f against control effort (w_a) and jerk smoothness (w_j): the jerk regulariser is what keeps the downstream flatness derivatives — the attitude quaternion and body rate of Section IV-B, obtained by differentiating the planned acceleration — well conditioned, so w_j couples the planner cost directly to the quality of the reference handed to the tracker. The NLP is formulated with CasADi [26] and solved offline with IPOPT [27].

IV. DIFFERENTIAL FLATNESS REFERENCE CONVERSION

A. FLAT OUTPUT AND DEPENDENCY STRUCTURE

The quadrotor is differentially flat with flat output $\boldsymbol{\sigma} = [\boldsymbol{p}^\top, \psi]^\top \in \mathbb{R}^4$, the position in $\mathcal{W}$ together with the yaw angle ψ [5], [23]. Flatness means that every state and input component of the body-rate model (1)–(4) can be expressed as an algebraic function of $\boldsymbol{\sigma}$ and a finite number of its time derivatives, without integration of the dynamics. For the body-rate model used here, the dependency chain reads $\boldsymbol{v} = \dot{\boldsymbol{p}}$; the specific force $\boldsymbol{f} = \ddot{\boldsymbol{p}} + g\,\boldsymbol{e}_3$ yields the collective thrust $T = m\,\|\boldsymbol{f}\|$ and the body z-axis $z_B = \boldsymbol{f}/\|\boldsymbol{f}\|$; the remaining two body axes are fixed by ψ, producing the rotation matrix $\mathbf{R}$ and hence the quaternion $\boldsymbol{q}$; and the body angular velocity $\boldsymbol{\omega}$ is the axial vector of $\mathbf{R}^\top\dot{\mathbf{R}}$. In this pipeline the pmm planner does not produce an independent yaw reference, so ψ is fixed by the downstream convention specified in Section IV-B. One departure from the closed-form flatness maps of [5], [23] is that the angular velocity $\boldsymbol{\omega}^*$ is computed via numerical forward finite difference of the discrete rotation sequence rather than analytically from position jerk; this choice is motivated in Section IV-B.

B. FROM PMM OUTPUTS TO FULL QUADROTOR REFERENCE

The pmm supplies position, velocity, and acceleration on the collocation grid; jerk is recovered by backward finite differences, $\boldsymbol{j}_k \approx (\boldsymbol{a}_k - \boldsymbol{a}_{k-1})/\Delta t^*$, and these translational quantities determine the remaining degrees of freedom of the quadrotor state through the chain laid out below. Newton's second law identifies the specific force $\boldsymbol{f}^* = \boldsymbol{a} + g\,\boldsymbol{e}_3$, from which the collective thrust and the body z-axis follow directly:

$$T^* = m\,\|\boldsymbol{f}^*\|, \qquad \boldsymbol{z}_B^* = \frac{\boldsymbol{f}^*}{\|\boldsymbol{f}^*\|}. \tag{9}$$

The remaining two body axes are fixed by the yaw convention $\psi^* = \operatorname{atan2}(v_y, v_x)$ with heading vector $\boldsymbol{x}_C = [\cos\psi^*, \sin\psi^*, 0]^\top$, giving

$$\boldsymbol{y}_B^* = \frac{\boldsymbol{z}_B^* \times \boldsymbol{x}_C}{\|\boldsymbol{z}_B^* \times \boldsymbol{x}_C\|}, \qquad \boldsymbol{x}_B^* = \boldsymbol{y}_B^* \times \boldsymbol{z}_B^*,$$
$$\mathbf{R}^* = \left[\,\boldsymbol{x}_B^* \mid \boldsymbol{y}_B^* \mid \boldsymbol{z}_B^*\,\right]. \tag{10}$$

The body z-axis in (9) is undefined only if the specific force vanishes, $\boldsymbol{a} = -g\boldsymbol{e}_3$, corresponding to pure free-fall; this configuration is never realised by the pmm output on the eight-gate circuit. The rotation matrix is converted to a unit quaternion with the Shepperd method [28], whose pivot $m = \arg\max\{1+\mathrm{tr}\mathbf{R}^*,\ 1+2R_{11}-\mathrm{tr},\ 1+2R_{22}-\mathrm{tr},\ 1+2R_{33}-\mathrm{tr}\}$ maximises numerical conditioning across the four branch formulas; to keep the discrete sequence on the same hemisphere of the double cover $\boldsymbol{q} \sim -\boldsymbol{q}$, a consecutive pair with negative inner product is flipped,

$$\boldsymbol{q}_k^{*\top}\boldsymbol{q}_{k-1}^* < 0 \;\Rightarrow\; \boldsymbol{q}_k^* \leftarrow -\boldsymbol{q}_k^*. \tag{11}$$

Differentiating $\mathbf{R}^*$ numerically then yields the body angular velocity. Because $\mathbf{R}^*$ is available only at the discrete collocation nodes, the identity $[\boldsymbol{\omega}^*]_\times = \mathbf{R}^{*\top}\dot{\mathbf{R}}^*$ is evaluated with a forward finite difference,

$$\dot{\mathbf{R}}_k^* \approx \frac{\mathbf{R}_{k+1}^* - \mathbf{R}_k^*}{\Delta t_k}, \tag{12}$$

and the angular velocity is extracted from $\boldsymbol{\Omega}_k = \mathbf{R}_k^{*\top}\dot{\mathbf{R}}_k^*$ via the vee map $\boldsymbol{\omega}_k^* = \boldsymbol{\Omega}_k^\vee$. The closed-form jerk-based angular-rate formula of Faessler *et al.* [23] is not applicable here because it requires continuous jerk, whereas the pmm delivers only a piecewise-constant acceleration on a non-uniform grid; for the same reason, finite differencing on a variable step Δt_k amplifies high-frequency content, which we attenuate in two steps. A causal 5-point uniform moving average, $\bar{\omega}_{i,k}^* = \frac{1}{5}\sum_{j=0}^{4}\omega_{i,k-j}^*$, first removes the differentiation noise while preserving the mean trend, and a subsequent hard saturation $|\bar{\omega}_i^*| \le \omega_{\max}^{\mathrm{ref}} < \omega_{\max}$ reserves a small headroom below the actuator limit for closed-loop corrections. The resulting full state reference and thrust feedforward

$$\boldsymbol{x}^* = \left[\boldsymbol{p}^{*\top}, \boldsymbol{v}^{*\top}, \boldsymbol{q}^{*\top}, \boldsymbol{\omega}^{*\top}\right]^\top, \qquad u_{\mathrm{ff}} = T^*, \tag{13}$$

are interpolated from the pmm collocation grid to the nmpc sampling grid using not-a-knot cubic splines, completing the algebraic bridge that an nmpc tracker can consume in full.

V. NMPC FORMULATION

A. STAGE AND TERMINAL COSTS

The nmpc receives reference information through a 17-component parameter vector,

$$\boldsymbol{p}_j = [\boldsymbol{p}_j^{*\top}, \boldsymbol{v}_j^{*\top}, \boldsymbol{q}_j^{*\top}, \boldsymbol{\omega}_j^{*\top}, T_j^*, \mathbf{0}_3^\top]^\top, \tag{14}$$

whose components are filled by the flatness bridge to the extent appropriate for each reference configuration (Section VI-B). Because the attitude component lives on $SO(3)$, the cost measures attitude error as a geodesic distance on the manifold rather than as a Euclidean difference in quaternion coordinates; a raw quaternion difference is discontinuous under the double cover $\boldsymbol{q} \sim -\boldsymbol{q}$ and destabilises the Gauss-Newton Hessian used by SQP-RTI. We adopt the logarithmic map on $SO(3)$ [29]. Denoting the body-frame error quaternion $\boldsymbol{q}_e = \boldsymbol{q}^{-1} \otimes \boldsymbol{q}^*$ with vector part $\boldsymbol{q}_{e,v} = [q_{e,x}, q_{e,y}, q_{e,z}]^\top$ and, to restrict the shortest-rotation branch, the sign convention $q_{e,w} \ge 0$ enforced by flipping $\boldsymbol{q}_e \leftarrow -\boldsymbol{q}_e$ when $q_{e,w} < 0$, the attitude error used in the cost is

$$\boldsymbol{e}_R = \log_{SO(3)}(\boldsymbol{q}_e) = \frac{2\operatorname{atan2}(\|\boldsymbol{q}_{e,v}\|,\, q_{e,w})}{\|\boldsymbol{q}_{e,v}\| + \varepsilon}\,\boldsymbol{q}_{e,v}, \tag{15}$$

with a fixed regularisation $\varepsilon = 10^{-9}$ that guarantees a finite Gauss-Newton Hessian at $\boldsymbol{q}^* = \boldsymbol{q}$ without branching, as required by the external-cost interface of acados.

The general stage cost penalises position, velocity, attitude, angular-rate, and input deviations from the reference,

$$\ell = \|\boldsymbol{p}^* - \boldsymbol{p}\|_{\mathbf{Q}_p}^2 + \|\boldsymbol{v}^* - \boldsymbol{v}\|_{\mathbf{Q}_v}^2 + \|\boldsymbol{e}_R\|_{\mathbf{Q}_q}^2$$
$$+ \|\boldsymbol{\omega}^* - \boldsymbol{\omega}\|_{\mathbf{Q}_\omega}^2 + \|\boldsymbol{u} - \boldsymbol{u}_{\mathrm{ff}}\|_{\mathbf{R}}^2, \tag{16}$$

with $\boldsymbol{e}_R$ from (15), diagonal positive-semidefinite weights $\mathbf{Q}_p, \mathbf{Q}_v, \mathbf{Q}_q, \mathbf{Q}_\omega, \mathbf{R}$, and control feedforward $\boldsymbol{u}_{\mathrm{ff}} = [T^*, 0, 0, 0]^\top$. The terminal cost V_f has the same form without the $\mathbf{R}$ term. Not all weight matrices are active in every configuration: setting $\mathbf{Q}_\omega = \mathbf{0}$ for nmpc-Att drops the angular-rate term entirely, which is the correct choice when no meaningful $\boldsymbol{\omega}^*$ reference is available, as detailed in Section VI-B. Numerical values of the non-zero weights appear in Section VI-A.

B. OPTIMAL CONTROL PROBLEM

At each sampling instant t_k the nmpc solves the finite-horizon problem

$$\min_{\boldsymbol{u}_{0:N-1}} \sum_{j=0}^{N-1} \ell(\boldsymbol{x}_j, \boldsymbol{u}_j, \boldsymbol{p}_j) + V_f(\boldsymbol{x}_N, \boldsymbol{p}_N), \tag{17a}$$

$$\text{s.t.} \quad \boldsymbol{x}_{j+1} = F_{\mathrm{ERK4}}(\boldsymbol{x}_j, \boldsymbol{u}_j), \quad j = 0, \dots, N-1, \tag{17b}$$

$$T_j \in [0, T_{\max}], \quad \left\|\boldsymbol{\omega}_j^c\right\|_\infty \le \omega_{\max}, \tag{17c}$$

$$\boldsymbol{x}_0 = \hat{\boldsymbol{x}}_k, \tag{17d}$$

where F_{ERK4} is an explicit RK4 discretisation of (1)–(4) over a control step h, N is the prediction horizon in steps, and $\boldsymbol{p}_j \in \mathbb{R}^{17}$ is the reference parameter vector of (14).

C. SOLVER CONFIGURATION

The OCP (17) is implemented in `acados` [14] with the SQP-RTI [15] algorithm (one QP per control step), a full-condensing HPIPM QP solver, an ERK4 integrator with 10 sub-steps per control interval, and a Gauss-Newton Hessian approximation. Because acados compiles the cost-weight matrices as constants into generated C code, each reference configuration requires a separately compiled solver; the two resulting binaries are otherwise structurally identical ($n_x = 13$, $n_u = 4$, $N = 50$). The initial nmpc call is warm-started from the flatness reference rather than from hover to reduce the transient phase.

VI. RESULTS

A. EXPERIMENTAL SETUP

The quadrotor is simulated in MuJoCo [30] with ground-truth state feedback; the internal integrator advances the rigid-body dynamics at $f_{\mathrm{sim}} = 500\,\mathrm{Hz}$, independently of the nmpc prediction model. Experiments run on a single core of an AMD Ryzen 7 7735HS (3.2 GHz, 16 GB RAM) under Ubuntu 22.04 with acados v0.5.3 and CasADi 3.7.2. A gate is counted as crossed when $\min_t \|\boldsymbol{p}(t) - \boldsymbol{g}_i\| < r$; G0 (launch position) is excluded, leaving seven evaluated gates.

Table 1 collects the platform and planner parameters introduced in Sections II–IV. The nmpc runs at $f_c = 100\,\mathrm{Hz}$ ($h = 10\,\mathrm{ms}$) with a prediction horizon of $N = 50$ steps ($T_h = 0.5\,\mathrm{s}$), an ERK4 integrator with 10 sub-steps per control interval, and an SQP-RTI convergence tolerance of $\epsilon_{\mathrm{tol}} = 10^{-4}$. These values follow standard practice and the real-time budget rather than a separate tuning procedure: $f_c = 100\,\mathrm{Hz}$ is the common control rate for body-rate agile nmpc [13], [14], and $N = 50$ ($T_h = 0.5\,\mathrm{s}$) is the largest horizon for which the present implementation keeps the P99 solve time within the 10 ms deadline on the test platform (Section VI-F); the resulting half-second look-ahead spans roughly one inter-gate transit ($T_f^*/N_g \approx 0.7\,\mathrm{s}$), enough to anticipate the upcoming gate without over-extending the horizon. The MuJoCo simulation runs at $f_{\mathrm{sim}} = 500\,\mathrm{Hz}$, 50 times faster than the controller, so that the physics integration is independent of the nmpc prediction model; Fig. 1 shows the simulation environment.

TABLE 1: Platform and Planner Parameters.

Parameter	Symbol	Value
Quadrotor platform (Section II)		
Drone mass	m	1.08 kg
Gravitational accel.	g	$9.81\,\mathrm{m\,s^{-2}}$
Rate-loop time const.	τ_{rc}	0.03 s
Max collective thrust	$T_{\max}$	42.4 N (TWR $\approx$ 4)
Max body angular rate	$\omega_{\max}$	$10.0\,\mathrm{rad\,s^{-1}}$
PMM planner / flatness bridge (Sec. III–IV)		
Collocation nodes	N_{PMM}	160
Number of gates	N_g	8
Gate radius	r	0.4 m
Reference rate clip	$\omega_{\max}^{ref}$	$9.0\,\mathrm{rad\,s^{-1}}$
MA filter window	—	5 samples

The position, velocity, and input weights are shared by both configurations: $\mathbf{Q}_p = \mathrm{diag}(50, 50, 60)$, $\mathbf{Q}_v = \mathrm{diag}(6, 6, 6)$, and $\mathbf{R} = \mathrm{diag}(0.3, 1, 1, 1)$. The attitude weight differs deliberately between the two: $\mathbf{Q}_q^{\mathrm{Full}} = \mathrm{diag}(15, 15, 15)$ drives the tracker to the complete tilt-corrected reference, whereas $\mathbf{Q}_q^{\mathrm{Att}} = \mathrm{diag}(1, 1, 1)$ is intentionally relaxed because nmpc-Att is fed a yaw-only quaternion that is geometrically inconsistent with the body tilt the aggressive pmm trajectory demands; a strong attitude penalty on a wrong reference would fight the horizontal thrust component and degrade the already-incomplete baseline further. nmpc-Full additionally activates $\mathbf{Q}_\omega = \mathrm{diag}(0.5, 0.5, 0.5)$, while $\mathbf{Q}_\omega = \mathbf{0}$ for nmpc-Att, which provides no angular-rate reference. Stage and terminal weights are equal.

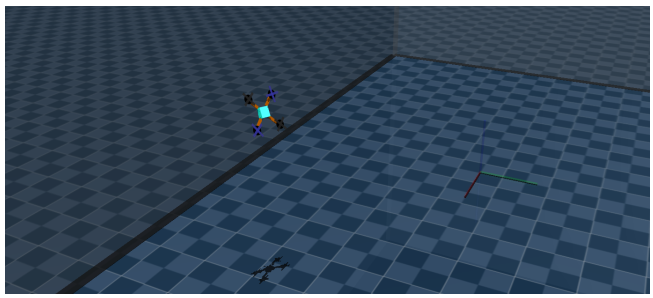

FIGURE 1: MuJoCo software-in-the-loop environment. The quadrotor (m = 1.08 kg) is simulated by MuJoCo's internal rigid-body integrator at 500 Hz while the body-rate nmpc runs at 100 Hz and communicates with the simulator over ROS 2. The gates (red rings) and the pmm-optimal reference are rendered for visual reference; a random external-force disturbance re-drawn at every reset provides trial-to-trial stochasticity.

B. REFERENCE CONFIGURATIONS

Both configurations track $\boldsymbol{p}^*(t)$ and $\boldsymbol{v}^*(t)$ from the pmm output and use identical dynamics, solver, and cost structure; the only variable is how much of the flatness reference is fed to the tracker. nmpc-Att uses a yaw-only quaternion $\boldsymbol{q}_\psi^* = [\cos(\psi^*/2), 0, 0, \sin(\psi^*/2)]^\top$ with no angular-rate or thrust feedforward ($\boldsymbol{\omega}^* = \mathbf{0}$, $T^* = 0$, $\mathbf{Q}_\omega = \mathbf{0}$). nmpc-Full applies the complete flatness chain (Section IV-B), adding the tilt-corrected quaternion, angular-rate reference, and thrust feedforward, with all cost terms active including $\mathbf{Q}_\omega = \mathrm{diag}(0.5, 0.5, 0.5)$.

C. EVALUATION CIRCUITS

We evaluate both controllers on two complementary reference circuits (Fig. 2, Table 2) chosen to probe opposite operating regimes of the flatness bridge.

Figure-8 (moderate racing). Eight gates form two interlocked lobes in the $\pm X$ direction with altitude alternation in $[1.0, 2.5]$ m. The starting gate is placed at the figure-8 crossover so that the drone's initial rest-to-speed transient aligns with the sustained flight direction, eliminating a cold-start artefact that would bias the comparison. The required body attitude remains inside the flat-yaw manifold ($z^*_{b,z} > 0$ throughout), so both configurations are expected to be physically capable of tracking the reference.

Vertical loop (acrobatic). Nine gates trace a *vertical* circular section in the *XZ* plane (radius $\approx$ 2 m, centred at $z = 3$ m) whose apex at $z = 5$ m forces an *inverted* body attitude. At the apex, the planned specific-force vector points downward in the world frame, so $z^*_{b,z} < 0$ and the tilt-corrected quaternion departs from the flat-yaw manifold. The yaw-only reference $\boldsymbol{q}^*_\psi = [\cos(\psi^*/2), 0, 0, \sin(\psi^*/2)]^\top$ used by nmpc-Att is *structurally* unable to represent any attitude with $z_{b,z} < 0$ (since $R^*_{33}(\boldsymbol{q}^*_\psi) = +1$ for all ψ^*), which turns this circuit into a feasibility test for the complete flatness inversion. The minimum observed $z^*_{b,z} = -0.97$ confirms the inversion geometrically (Fig. 2(b)).

TABLE 2: PMM Solutions for the Two Evaluation Circuits.

	Figure-8	Vertical Loop
Gates (N_g)	8	9
Arc length [m]	54.8	31.9
T^*_f [s]	5.55	5.40
v_{mean} [m s^{-1}]	9.85	5.89
v_{peak} [m s^{-1}]	16.55	12.10
$z^*_{b,z,\mathrm{min}}$	-0.34	-0.97 (inverted)
$\max \lvert\boldsymbol{\omega}^*\rvert$ [rad s^{-1}]	9.00 (clipped)	9.00 (clipped)
T^* range [N]	[5.8, 42.4]	[1.5, 42.4]

Figure 3 confirms that the flatness-derived feedforward is non-trivial on the acrobatic circuit: angular rates reach the $\pm 9\,\mathrm{rad\,s^{-1}}$ saturation clip in the loop and collective thrust drops to $\approx 3.3\,\mathrm{N}$ at the inverted apex (near the theoretical minimum for a circular orbit at $v^2/R \approx g$). Both signals are identically zero in nmpc-Att.

D. TRACKING PERFORMANCE

We run $N = 25$ software-in-the-loop (SiL) trials per controller per circuit against the MuJoCo physics engine driven through ROS 2. Stochasticity arises from the randomly seeded external-force disturbance that MuJoCo re-draws at every scene reset, so each of the 100 total trials is an independent realisation. Trials in which either controller diverged (RMSE $> 5\,\mathrm{m}$, attributable to the tail of the disturbance distribution) are flagged as *blow-ups* and excluded from the statistics under a paired rule (a trial is dropped if *either* controller diverges, so both are evaluated on the same disturbance realisations), leaving $23/25$ valid trials on the figure-8 and $18/25$ on the more demanding vertical loop. We report the gate-crossing rate alongside the continuous metrics. Table 3 consolidates all four configurations with mean $\pm$ standard deviation over valid trials.

TABLE 3: SiL Tracking Performance: Common-Practice Yaw Reference (nmpc-Att) vs. Complete Flatness Inversion (nmpc-Full) on Both Circuits ($N = 25$ MuJoCo trials with random external-force disturbance; mean $\pm$ std over valid trials; Gates: mean crossings per trial; bold = best).

Circ.	Ctrl.	RMSE [m]	Gates	d_{gate} [m]	P99 [ms]
Fig8	Att	0.407 ± 0.007	$3.09/7$	0.373 ± 0.004	4.39
	Full	$\mathbf{0.308 \pm 0.007}$	$\mathbf{7/7}$	$\mathbf{0.146 \pm 0.004}$	3.86
Loop	Att	0.289 ± 0.007	$5/8$	0.300 ± 0.003	3.86
	Full	$\mathbf{0.211 \pm 0.009}$	$\mathbf{8/8}$	$\mathbf{0.151 \pm 0.005}$	3.85

1) Figure-8 Circuit

On the seven-gate figure-8 (peak speed $16.6\,\mathrm{m\,s^{-1}}$, peak acceleration $36.3\,\mathrm{m\,s^{-2}}$), nmpc-Full completes all seven gates in every valid trial ($7/7$), while nmpc-Att crosses only $3.09/7$ on average ($44.1\,\%$). nmpc-Full reduces position RMSE by $24.3\,\%$ ($0.407\,\mathrm{m} \to 0.308\,\mathrm{m}$) and mean gate-crossing distance by $60.8\,\%$ ($0.373\,\mathrm{m} \to 0.146\,\mathrm{m}$). Figure 2(a) and Fig. 4 visualise the gap as a consistent offset concentrated around gate transitions, produced by the yaw-only attitude reference fighting the horizontal-thrust tilt the aggressive PMM trajectory demands.

2) Vertical Loop

On the eight-gate loop (peak speed $12.1\,\mathrm{m\,s^{-1}}$, peak acceleration $33.8\,\mathrm{m\,s^{-2}}$), the categorical separation is even sharper: nmpc-Full completes $8/8$ gates in every valid trial, whereas nmpc-Att averages only $5/8$ ($62.5\,\%$); the missed gates cluster around the inverted apex (G2–G5), which the yaw-only reference cannot represent by construction ($R^*_{33}(\boldsymbol{q}^*_\psi) = +1$ for all ψ^*). nmpc-Full reduces position RMSE by $27.2\,\%$ ($0.289\,\mathrm{m} \to 0.211\,\mathrm{m}$) and mean gate-crossing distance by $49.6\,\%$ ($0.300\,\mathrm{m} \to 0.151\,\mathrm{m}$). The attitude-feasibility argument is structural: no choice of cost weights can drive nmpc-Att to an attitude with $z_{b,z} < 0$, so the deficit is inherent to the reference, not an artefact of tuning. Figure 5 shows that nmpc-Att exceeds the gate radius precisely at the inverted section, while nmpc-Full stays inside at every gate. Together, the two circuits isolate *what gains can fix* (figure-8, a quantitative gap) from *what only reference enrichment can fix* (loop, a categorical gap).

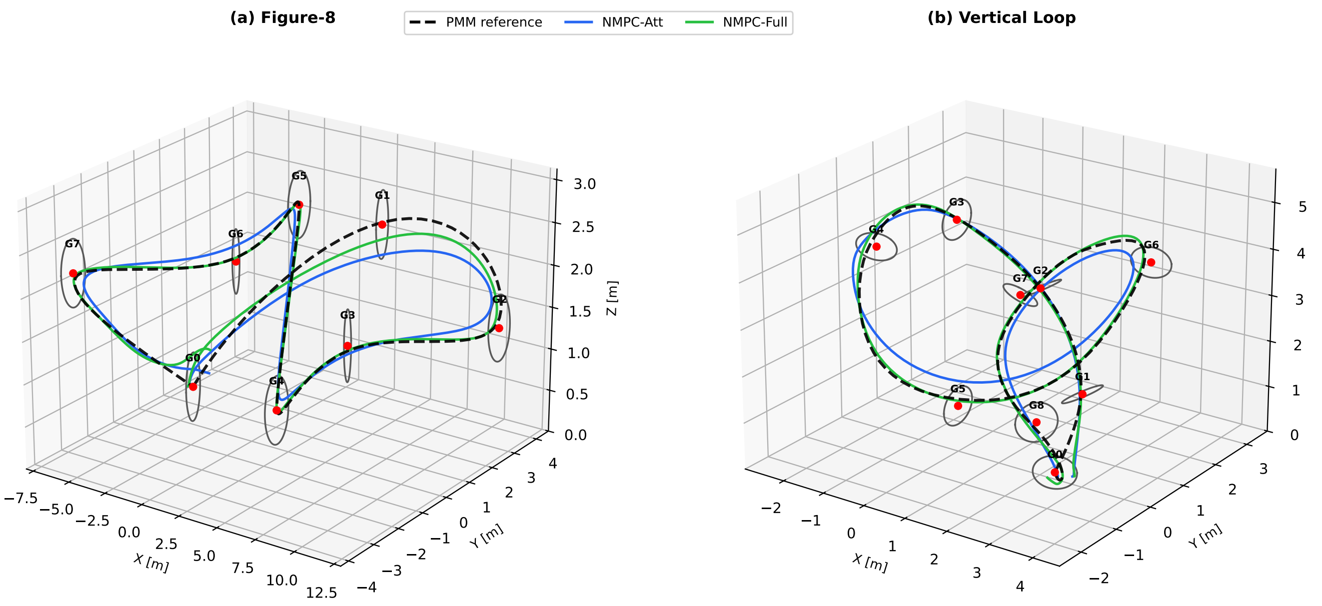

FIGURE 2: Evaluation circuits in 3-D. (a) Figure-8: moderate racing reference with altitude alternation (crossover start eliminates the cold-start transient). (b) Vertical loop: acrobatic reference whose apex (G3) forces an inverted body attitude ($z^*_{b,z} < 0$), exceeding the representational range of any yaw-only attitude reference. Black dashed line: pmm reference; blue: nmpc-Att; green: nmpc-Full; red markers: gate centres.

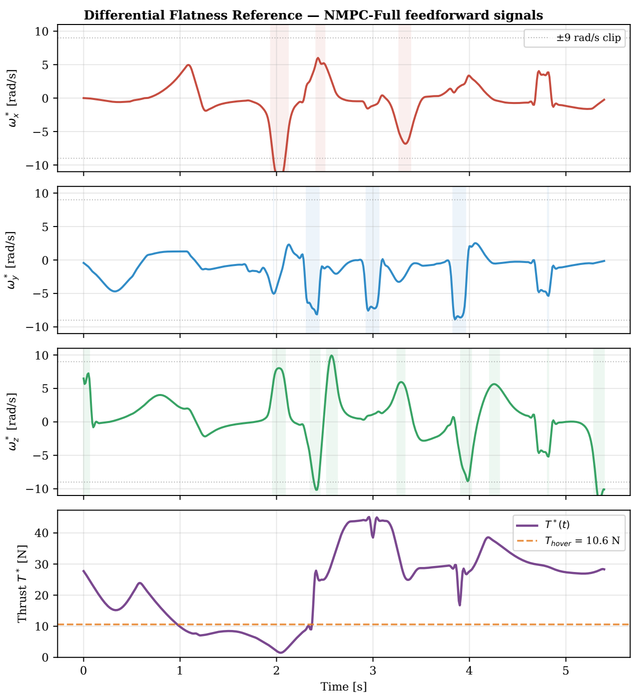

FIGURE 3: Flatness-derived feedforward on the vertical loop. *Top three panels*: body angular-rate reference $\boldsymbol{\omega}^*(t)$ per axis (shaded: $|\omega^*_i| > 5\,\mathrm{rad\,s^{-1}}$; dotted: $\pm 9\,\mathrm{rad\,s^{-1}}$ clip). *Bottom*: collective thrust $T^*(t)$ (purple) versus hover thrust $T_{\mathrm{hover}} = mg = 10.6\,\mathrm{N}$. Thrust collapses to $\approx 3.3\,\mathrm{N}$ at the inverted apex. All four signals are zero in nmpc-Att.

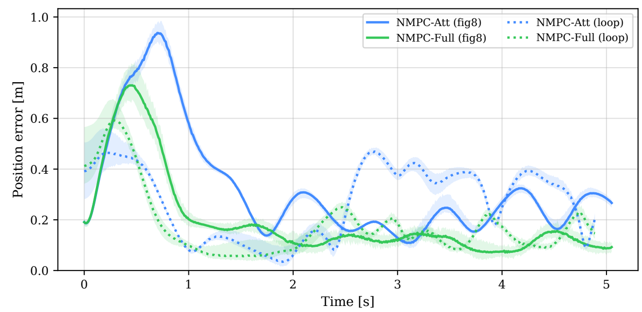

FIGURE 4: Position error norm $\|\boldsymbol{p}(t) - \boldsymbol{p}^*(t)\|$ on both circuits (fig-8: solid; loop: dashed). In the figure-8, nmpc-Full stays consistently below nmpc-Att with peaks at gate-crossing instants. In the vertical loop, nmpc-Att diverges during the inverted-apex traversal ($t \approx 2.5$–4.5 s); nmpc-Full tracks the reference throughout.

E. MECHANISM: ATTITUDE AND RATE TRACKING

Figure 6 resolves the mechanism at the attitude level. The nmpc-Att attitude error exceeds 90° during the inverted section because the reference it is following ($\boldsymbol{q}^*_\psi$) is geometrically the wrong attitude — the target is at the antipode of the feasible tilt, and no amount of feedback gain can recover it. nmpc-Full receives the correct tilt-corrected reference and its error stays bounded.

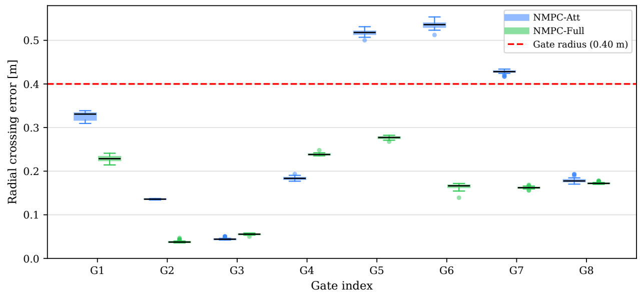

FIGURE 5: Vertical loop: minimum 3-D distance to each gate centre, aggregated as a box plot over the 18 valid SiL trials per controller. Dashed line: gate radius $r = 0.4\,\mathrm{m}$. nmpc-Att exceeds the radius at gates G2–G5 (inverted section); nmpc-Full remains below r at every gate.

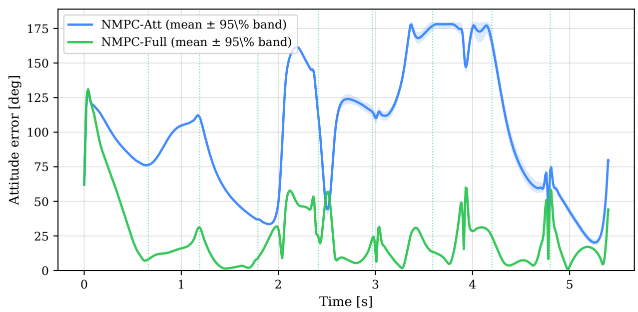

FIGURE 6: Vertical loop: attitude error $\|\log(\mathbf{R}\mathbf{R}^{*\top})\|_F$ in degrees. nmpc-Att diverges at the apex because $\boldsymbol{q}^*_\psi$ cannot represent the inverted attitude; nmpc-Full tracks the flatness-derived reference with bounded error.

F. COMPUTATIONAL PERFORMANCE

Reference enrichment carries *no* computational penalty. Both configurations share identical OCP dimensions and differ only in weight values and parameter content, so solver work per step is invariant. Across both circuits and all 82 valid SiL trials, P99 solve time stays below $4.4\,\mathrm{ms}$, giving $>2\times$ margin to the $10\,\mathrm{ms}$ nmpc deadline (Table 3, last column). Hence the categorical feasibility gain on the loop and the $24\,\%$ to $27\,\%$ RMSE reduction on both circuits are delivered with zero trade-off against real-time budget.

G. DISCUSSION

The two circuits split the contribution of the full flatness inversion into two distinct claims.

Quantitative, in the flat-yaw regime. On the figure-8, where the yaw-only reference is at least geometrically feasible, nmpc-Full reduces position RMSE by $24.3\,\%$ and more than halves the mean gate-crossing distance ($60.8\,\%$), at identical solver cost. This isolates the value of the angular-rate and thrust feedforward: without them, the tracker must infer the required body tilt from error feedback and absorbs a controller-cycle lag at every acceleration transition, which at $16.6\,\mathrm{m\,s^{-1}}$ peak speed pushes the mean crossing distance beyond the gate radius and collapses the crossing rate of nmpc-Att to $44.1\,\%$.

Categorical, in the non-flat regime. On the vertical loop, the yaw-only reference is geometrically *wrong* — its third rotation column is always $+e_3$, while the circuit requires $-e_3$ at the apex. No choice of cost weights can fix this; only reference enrichment does. This is the stronger answer to the interface question of Song *et al.* [4]: the limiting factor of the *planner–controller interface* is not its modular architecture but the *representational completeness* of the reference passed across it. Enriching the interface with the flatness output closes both the quantitative and the categorical gap with no additional compute.

Positioned against the cited literature, the proposed pipeline occupies a distinct point in the design space. Foehn *et al.* [1] pair a pmm planner with MPCC tracking but do not feed a flatness-derived full-state reference to the tracker; we retain their fast point-mass planning while passing the complete flat output $(\boldsymbol{p}, \boldsymbol{v}, \boldsymbol{q}, \boldsymbol{\omega}, T)$ to the nmpc. The MPCC formulations of Romero *et al.* [2] and Krinner *et al.* [21] reach excellent crossing rates but fuse planning and tracking into a single optimisation, so the planner and controller cannot be designed, validated, or replaced independently; our architecture keeps the two stages modular while recovering the tracking completeness that the fused formulation achieves implicitly. Finally, Sun *et al.* [19] benchmark nmpc against differential-flatness-based control over 76 trajectories but hold the reference structure fixed, whereas the present study *ablates* that structure and isolates which flat components (angular rate, tilt-corrected attitude) drive gate-crossing performance. The result is a pipeline with the modularity of decomposed approaches and the reference completeness usually exclusive to fused ones, at no additional solver cost.

Two limitations qualify the scope. The SiL experiment includes MuJoCo's random external-force disturbance (re-drawn at every reset) and round-trip ROS 2 messaging, but does not cover communication latency beyond the ROS 2 loopback or aerodynamic effects such as rotor wash and drag, leaving full sim-to-real robustness unquantified. A learning-augmented controller that infers the flatness feedforward from rollouts could in principle bridge the figure-8 gap, but the loop result suggests that *reference completeness*, not controller capacity, is the binding constraint for acrobatic flight.

VII. CONCLUSION

This paper presented a hierarchical planning and control framework for time-optimal agile gate navigation with quadrotor UAVs. A Point Mass Model planner generates a time-optimal reference trajectory offline; a differential flatness bridge maps this reference to the full quadrotor state; and an SQP-RTI nmpc tracks the converted reference in real time at $100\,\mathrm{Hz}$. Two reference configurations were evaluated in MuJoCo physics-engine simulation to isolate the value of the flatness inversion. nmpc-Att, representing common practice, derives a yaw-only quaternion from the velocity direction and includes no angular-rate or thrust feedforward;

its cost correctly omits the angular-rate penalty to avoid contradictory objectives. nmpc-Full applies the complete flatness chain, providing the tilt-corrected quaternion, angular-rate, and thrust feedforward with all cost terms active. Validation was performed in a software-in-the-loop setup driving MuJoCo through ROS 2 with a random external-force disturbance re-drawn at every reset; $N = 25$ trials were run per circuit per controller under a consistent $4\,\mathrm{g}$ thrust envelope shared by planner and tracker. On the seven-gate figure-8 (peak speed $16.6\,\mathrm{m\,s^{-1}}$) and the eight-gate vertical loop (peak speed $12.1\,\mathrm{m\,s^{-1}}$), nmpc-Full achieves $100\,\%$ gate-crossing in every valid trial, while nmpc-Att completes only $44.1\,\%$ (fig-8) and $62.5\,\%$ (loop). nmpc-Full reduces position RMSE by $24.3\,\%$ (fig-8) and $27.2\,\%$ (loop) and mean gate-crossing distance by $60.8\,\%$ and $49.6\,\%$ respectively, while the solver P99 time remains below $4.4\,\mathrm{ms}$ for both configurations. The figure-8 gap is quantitative — driven by the missing angular-rate and thrust feedforward at sharp acceleration transitions — whereas the loop gap is categorical: the yaw-only reference cannot represent the inverted attitude at the apex, so only reference enrichment closes it. Ongoing work extends the validation to hardware-in-the-loop with realistic communication latencies and aerodynamic effects, and to deployment on a physical quadrotor platform to quantify the sim-to-real gap.

CODE AVAILABILITY

The source code and experiment scripts are openly available at https://github.com/bryansgue/Time_optimal_planing-NMPC.

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REFERENCES

[1] P. Foehn, A. Romero, and D. Scaramuzza, "Time-optimal planning for quadrotor waypoint flight," *Sci. Robot.*, vol. 6, no. 56, p. eabh1221, 2021.
[2] A. Romero, S. Sun, P. Foehn, and D. Scaramuzza, "Model predictive contouring control for time-optimal quadrotor flight," *IEEE Trans. Robot.*, vol. 38, no. 6, pp. 3340–3356, 2022.
[3] E. Kaufmann, L. Bauersfeld, A. Loquercio, M. Müller, V. Koltun, and D. Scaramuzza, "Champion-level drone racing using deep reinforcement learning," *Nature*, vol. 620, no. 7976, pp. 982–987, 2023.
[4] Y. Song, A. Romero, M. Müller, V. Koltun, and D. Scaramuzza, "Reaching the limit in autonomous racing: Optimal control versus reinforcement learning," *Sci. Robot.*, vol. 8, no. 82, p. eadg1462, 2023.
[5] D. Mellinger and V. Kumar, "Minimum snap trajectory generation and control for quadrotors," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2011, pp. 2520–2525.
[6] C. Richter, A. Bry, and N. Roy, "Polynomial trajectory planning for aggressive quadrotor flight in dense indoor environments," in *Robotics Research (ISRR)*, 2016, pp. 649–666.
[7] D. Mellinger, N. Michael, and V. Kumar, "Trajectory generation and control for precise aggressive maneuvers with quadrotors," *Int. J. Robot. Res.*, vol. 31, no. 5, pp. 664–674, 2012.
[8] M. W. Mueller, M. Hehn, and R. D'Andrea, "A computationally efficient motion primitive for quadrocopter trajectory generation," *IEEE Trans. Robot.*, vol. 31, no. 6, pp. 1294–1310, 2015.
[9] C. Qin, M. S. J. Michet, J. Chen, and H. H.-T. Liu, "Time-optimal gate-traversing planner for autonomous drone racing," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2024, pp. 14 513–14 519, best Paper Award on Unmanned Aerial Vehicles.
[10] Z. Zhou, G. Wang, J. Sun, J. Wang, and J. Chen, "Efficient and robust time-optimal trajectory planning and control for agile quadrotor flight," *IEEE Robotics and Automation Letters*, 2023.
[11] R. Penicka, Y. Song, E. Kaufmann, and D. Scaramuzza, "Minimum-time quadrotor waypoint flight in cluttered environments," in *IEEE Robot. Autom. Lett.*, vol. 7, no. 2, 2022, pp. 5719–5726.
[12] A. Loquercio, E. Kaufmann, R. Ranftl, M. Müller, V. Koltun, and D. Scaramuzza, "Learning high-speed flight in the wild," *Sci. Robot.*, vol. 6, no. 59, p. eabg5810, 2021.
[13] D. Hanover, P. Foehn, S. Sun, E. Kaufmann, and D. Scaramuzza, "Performance, precision, and payloads: Adaptive nonlinear MPC for quadrotors," *IEEE Robot. Autom. Lett.*, vol. 7, no. 2, pp. 690–697, 2022.
[14] R. Verschueren, G. Frison, D. Kouzoupis, J. Frey, N. van Duijkeren, A. Zanelli, B. Novoselnik, T. Albin, R. Quirynen, and M. Diehl, "acados — a modular open-source framework for fast embedded optimal control," *Math. Program. Comput.*, vol. 14, no. 1, pp. 147–183, 2022.
[15] M. Diehl, H. G. Bock, and J. P. Schlöder, "A real-time iteration scheme for nonlinear optimization in optimal feedback control," *SIAM J. Control Optim.*, vol. 43, no. 5, pp. 1714–1736, 2005.
[16] G. Torrente, E. Kaufmann, P. Foehn, and D. Scaramuzza, "Data-driven MPC for quadrotors," *IEEE Robot. Autom. Lett.*, vol. 6, no. 2, pp. 3769–3776, 2021.
[17] E. Tal and S. Karaman, "Accurate tracking of aggressive quadrotor trajectories using incremental nonlinear dynamic inversion and differential flatness," in *IEEE Trans. Control Syst. Technol.*, vol. 29, no. 3, 2021, pp. 1203–1218.
[18] M. Kamel, T. Stastny, K. Alexis, and R. Siegwart, "Model predictive control for trajectory tracking of unmanned aerial vehicles using ROS," in *Robot Operating System (ROS): The Complete Reference (Volume 2)*, 2017, pp. 3–39.
[19] S. Sun, A. Romero, P. Foehn, E. Kaufmann, and D. Scaramuzza, "A comparative study of nonlinear MPC and differential-flatness-based control for quadrotor agile flight," *IEEE Trans. Robot.*, vol. 38, no. 6, pp. 3357–3373, 2022.
[20] A. Romero, R. Penicka, and D. Scaramuzza, "Time-optimal online replanning for agile quadrotor flight," *IEEE Robot. Autom. Lett.*, vol. 7, no. 3, pp. 7730–7737, 2022.
[21] S. Krinner, A. Romero, L. Bauersfeld, and D. Scaramuzza, "MPCC++: Model predictive contouring control for time-optimal flight with safety constraints," *IEEE Robot. Autom. Lett.*, 2024.
[22] D. Falanga, S. Kim, and D. Scaramuzza, "How fast is too fast? The role of perception latency in high-speed sense and avoid," in *IEEE Robot. Autom. Lett.*, vol. 4, no. 2, 2019, pp. 1884–1891.
[23] M. Faessler, A. Franchi, and D. Scaramuzza, "Differential flatness of quadrotor dynamics subject to rotor drag for accurate tracking of high-speed trajectories," *IEEE Robot. Autom. Lett.*, vol. 3, no. 2, pp. 620–626, 2018.
[24] M. Faessler, D. Falanga, and D. Scaramuzza, "Thrust mixing, saturation, and body-rate control for accurate aggressive quadrotor flight," *IEEE Robot. Autom. Lett.*, vol. 2, no. 2, pp. 561–568, 2017.
[25] M. Greeff and A. P. Schoellig, "Flatness-based model predictive control for quadrotor trajectory tracking," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, 2018, pp. 6740–6745.
[26] J. A. E. Andersson, J. Gillis, G. Horn, J. B. Rawlings, and M. Diehl, "CasADi: A software framework for nonlinear optimization and optimal control," *Math. Program. Comput.*, vol. 11, no. 1, pp. 1–36, 2019.
[27] A. Wächter and L. T. Biegler, "On the implementation of an interior-point filter line-search algorithm for large-scale nonlinear programming," *Math. Program.*, vol. 106, no. 1, pp. 25–57, 2006.
[28] S. W. Shepperd, "Quaternion from rotation matrix," *J. Guid. Control*, vol. 1, no. 3, pp. 223–224, 1978.
[29] T. Lee, M. Leok, and N. H. McClamroch, "Geometric tracking control of a quadrotor UAV on SE(3)," in *Proc. IEEE Conf. Decis. Control (CDC)*, 2010, pp. 5765–5770.
[30] E. Todorov, T. Erez, and Y. Tassa, "MuJoCo: A physics engine for model-based control," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, 2012, pp. 5026–5033.

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Fig/alex.png

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Fig/Jose.png

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